Direct Sums.

Prop. (18.6.1) Let V, W be vect. sp. over K. Then the cartesian product $V \times W$ becomes a vect. space over K if we endow it with the following operations:

* $(v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2) \quad \forall\ (v_1, w_1), (v_2, w_2) \in V \times W.$
* $\alpha \cdot (v, w) := (\alpha v, \alpha w) \quad \forall\ \alpha \in K,\ (v, w) \in V \times W.$
* $0_{V \times W} := (0_V, 0_W)$ (the zero, or neutral element, of $V \times W$)

This new vect. space $V \times W$ is usually denoted $V \oplus W$ and is called the direct sum of V and W. We write the elements of $V \oplus W$ as (v, w) or $v \oplus w$.

Proof of Prop. Exc. Check that all the axioms of Vect. sp. are satisfied. ∎

Remark. $V \oplus W$ comes together with the following canonical maps:

1) $i_V : V \longrightarrow V \oplus W, \quad i_V(v) := (v, 0).$
2) $i_W : W \longrightarrow V \oplus W, \quad i_W(w) := (0, w).$

} Canonical embeddings.

3) $p_V : V \oplus W \longrightarrow V, \quad p_V(v, w) := v.$
4) $p_W : V \oplus W \longrightarrow W, \quad p_W(v, w) := w.$

} Canonical projections.

These maps are all linear (check!). The maps p_V, p_W are surjective, and i_V, i_W are injective (check!). Because of the maps i_V, i_W we can sometimes view V and W as subspaces of $V \oplus W$,

$V \subseteq V \oplus W, \quad W \subseteq V \oplus W.$

$V \ni v \longleftrightarrow (v, 0) \in V \oplus W,$
$W \ni w \longleftrightarrow (0, w) \in V \oplus W.$

Prop. (18.b.2) Let V be a vect. sp. over K, $\mathcal{U} \subseteq V$ a subspace and $\mathcal{U}' \subseteq V$ another subspace which is a complement of $\mathcal{U}$ in V (recall: this means $\mathcal{U}+\mathcal{U}'=V$ and $\mathcal{U}\cap\mathcal{U}'=\{0\}$). Then the map

$$\varphi: \mathcal{U}\oplus\mathcal{U}' \longrightarrow V, \qquad \varphi(u,u') := u+u' \quad \text{is a } \underbrace{\text{linear isomorphism}}_{\text{(meaning: the map is linear and an iso.)}}.$$

Proof. Linearity of φ follows directly from the definitions. (Check!).

Injectivity of φ: Suppose $\varphi(u,u')=0. \Rightarrow u+u'=0. \Rightarrow u=-u'$.

But $\mathcal{U}'$ is a complement of $\mathcal{U}$, hence $\mathcal{U}\cap\mathcal{U}'=\{0\}$ (by definition).

So $\underset{\mathcal{U}}{\underset{\cap}{u}} = -\underset{\mathcal{U}'}{\underset{\cap}{u'}}$ implies that $u=u'=0. \Rightarrow (u,u')=(0,0)$. This proves that $\mathrm{Ker}(\varphi)=\{0_{\mathcal{U}\oplus\mathcal{U}'}\}$, hence φ is injective.

Surjectivity of φ: Since $\mathcal{U}'$ is a complement of $\mathcal{U}$ in V, we have, by def., that $\mathcal{U}+\mathcal{U}'=V$. Let $v\in V$. $\exists\ u\in\mathcal{U}, u'\in\mathcal{U}'$ s.t. $u+u'=v. \Rightarrow \varphi(u,u')=v$. This shows $\mathrm{Im}(\varphi)=V$, i.e. φ is surjective. ∎

Remark. When $\mathcal{U}'$ is a complement of $\mathcal{U}$ in V, then

$\mathcal{U}+\mathcal{U}'=V$ and we also have a canonical iso. $\mathcal{U}\oplus\mathcal{U}' \xrightarrow[\varphi]{\cong} V$

But formally speaking $\mathcal{U}\oplus\mathcal{U}'$ is a different vect. sp. than V.

Generalization to more summands: Let $\mathcal{U}_1,\dots,\mathcal{U}_m$ be finitely many vect. spaces over K. We can define $\mathcal{U}_1\oplus\dots\oplus\mathcal{U}_m := \mathcal{U}_1\times\dots\times\mathcal{U}_m$, with the coordinatwise / componentwise operations. For infinite families $\mathcal{U}_i$, $i\in I$, one can define $\bigoplus_{i\in I}\mathcal{U}_i = \left\{ t: I\longrightarrow \bigcup_{i\in I}\mathcal{U}_i \;\middle|\; \begin{array}{l} t(i)\in\mathcal{U}_i\ \forall i\in I,\ t(i)\neq 0 \text{ for} \\ \text{at most finitely many } i\text{'s} \end{array} \right\}$.